Practical - 15

Practical Title: Final Integration, Testing, Documentation, GitHub Submission, and Local Deployment

Problem definition: Integrate all modules into a single deployable StudentHub portal. Verify navigation, responsiveness, authentication, database operations, security checks, AJAX functionality, and documentation. Deploy locally using XAMPP/Apache virtual host and submit GitHub repository.

Theory:

- Are all pages properly linked and navigable?

Yes, all required pages are linked through the navigation menu, and each page opens correctly without broken links.

- Are sessions, roles, validations, and database operations working end-to-end?

Yes, after testing, registration, login, logout, session management, role-based access, form validation, and database operations work correctly.

- Are security and accessibility checklists completed?

The application is checked for password hashing, prepared statements, session protection, input validation, image alt text, form labels, keyboard navigation, and readable contrast.

- Is project documentation sufficient for installation and testing?

Yes, the README explains the technologies, installation steps, database setup, features, and testing process. The project also includes an SQL dump and test report.

- Is the project repository clean and properly committed?

Yes, the required project files are committed and pushed to GitHub, and temporary files and sensitive credentials are excluded.

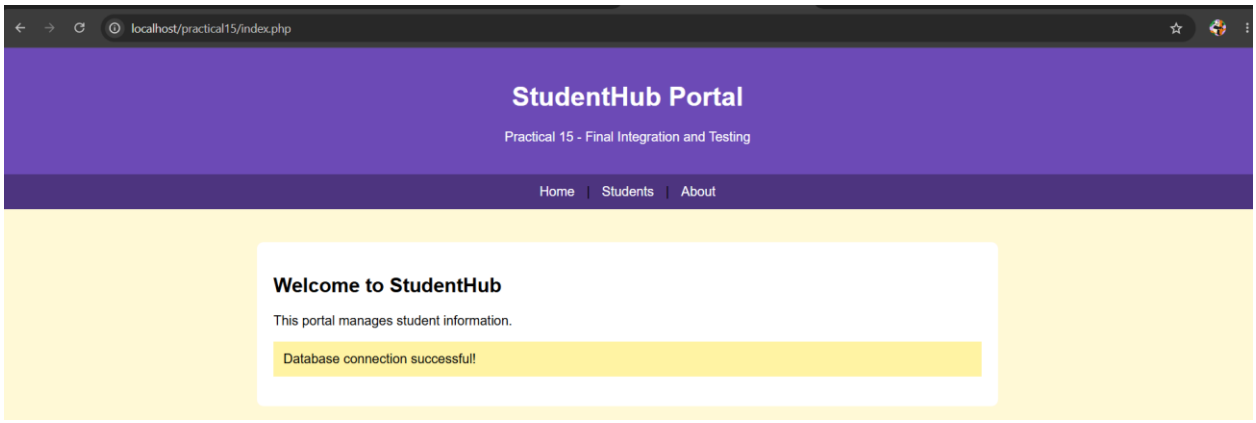
A. Implementation/ Output/Result **(Keep only one option which is suitable to practical)**

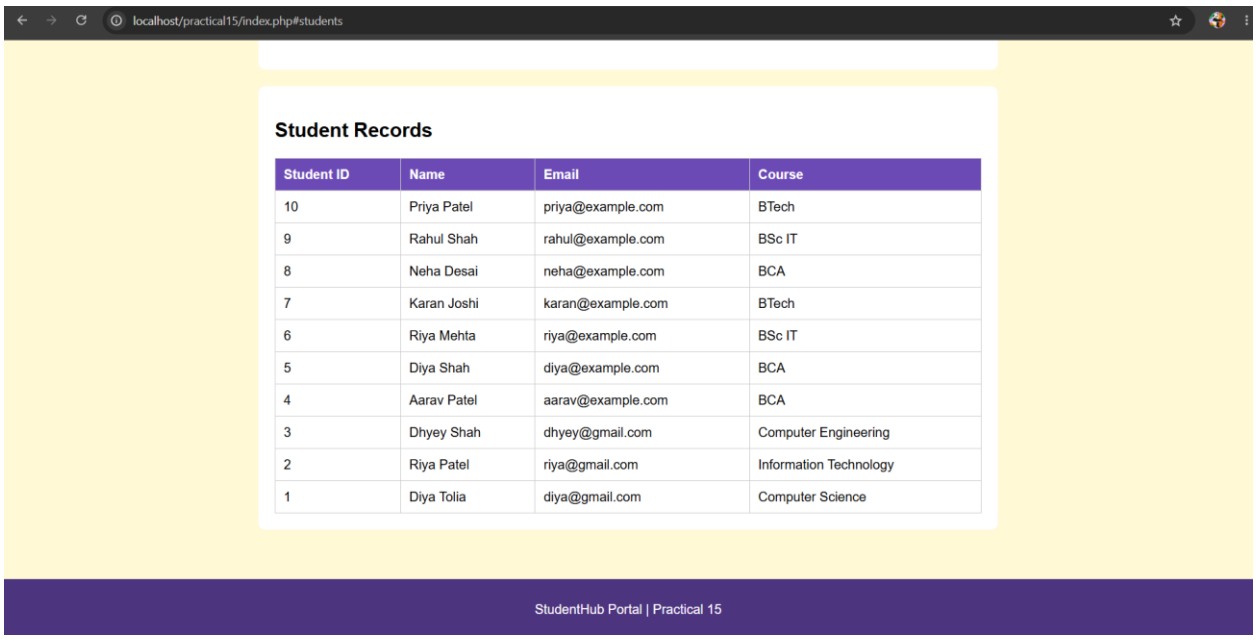
2. PHP syntax and functions used:

PHP element	Purpose

`require_once`	Includes the database connection file only once.
`new mysqli()`	Connects PHP to the MySQL database.
`if` and `else`	Check conditions and display appropriate output.
`while`	Repeats code to display each student record.
`fetch_assoc()`	Retrieves a database row as an associative array.
`htmlspecialchars()`	Helps prevent HTML injection when displaying database values.
`close()`	Closes the database connection.
`$_SERVER`	Provides server and request information when required.

A.Evidence of Work (Screenshots)





B.GitHub Repository Link:

https://github.com/diyatolia1212/StudentHub/tree/42d637fdc457c7937b304074f48efb4c8b275c89/practical15

C.Challenges Faced:

- Broken page links were corrected by checking file paths.
- Database connection errors were resolved by checking database credentials and table names.
- Login issues were fixed by verifying sessions and role checks.

Learning Outcomes:

After completing this practical, I learned how to integrate different web development modules into a single application.

Faculty Remarks: **(Keep this blank for faculty)**